# METHOD AND SYSTEM FOR PIXEL-BASED NETWORK ROUTING IN A MASSIVELY PARALLEL OPTICAL INTERCONNECT

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**Title:** METHOD AND SYSTEM FOR PIXEL-BASED NETWORK ROUTING IN A MASSIVELY PARALLEL OPTICAL INTERCONNECT

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## Abstract

A method and system for software-defined network routing and multiplexing over a massively parallel optical interconnect. The method treats each channel in a two-dimensional emitter array (e.g., each pixel in a micro-display) as an addressable data conduit. Routing information is encoded by mapping data packets to specific emission characteristics of the channels. For example, data intended for different network destinations can be assigned to different wavelengths (colors). A processor at the transmitting end acts as a network switch, analyzing incoming data and generating a frame where the color of each pixel determines the destination of the data it carries. A receiver decodes the frame and routes the data from each pixel based on its color. This technique allows a single physical interconnect to simultaneously carry multiple, logically separated data streams to different destinations, creating a high-density, software-defined network fabric.

## Background of the Invention

In high-performance computing, datacenters, and telecommunications, network routing is traditionally handled by dedicated hardware switches and routers. These devices physically direct data packets from an input port to an output port based on address information in the packet headers. While effective, this approach requires complex and expensive hardware, and the total bandwidth is limited by the number of physical ports on the switch.

As data transmission moves towards massively parallel optical interconnects capable of terabit-per-second speeds, traditional routing hardware becomes a significant bottleneck. There is a need for a new routing paradigm that can operate at the speed and scale of these next-generation interconnects and that allows for more flexible, software-defined control over data flows. The present invention addresses this by moving the routing logic from a centralized hardware switch to the transmission protocol of the optical interconnect itself.

## Summary of the Invention

The present invention is a method for network routing that uses the emission characteristics of channels in a massively parallel optical interconnect to encode destination information. The invention leverages an interconnect system comprising a two-dimensional array of emitters (e.g., pixels in a display) and a corresponding array of sensors.

The core of the invention is a process where a processor, acting as a network switch, receives multiple incoming data streams intended for different destinations. Instead of sending these streams to different physical ports, it multiplexes them onto a single optical interconnect. It achieves this by assigning a unique emission characteristic, such as a specific wavelength (color), to each destination network.

The processor generates a sequence of data frames for the emitter array. For each frame, it analyzes the incoming data packets, determines their destination, and places the data into a pixel, setting that pixel's color to correspond to the assigned destination. For example, all data for "Network A" is encoded in red pixels, "Network B" in green pixels, and "Network C" in blue pixels.

The entire frame, containing a mix of red, green, and blue pixels, is transmitted in parallel. At the receiving end, a decoder analyzes the frame, reads the color of each pixel,

and routes the data contained within that pixel to the appropriate destination network. This creates multiple, logically separate, high-bandwidth data streams over a single physical cable, a form of ultra-high-density, software-defined wavelength-division multiplexing.

## Statement of License Grant to the United States Government

The inventor, Reinaldo Javier Aguiar Marcano, hereby grants to the United States Government and its agencies a perpetual, royalty-free, non-exclusive license to practice and use the invention described herein for any and all governmental and non-commercial purposes.

## Brief Description of the Drawings

**FIG. 1** is a conceptual diagram illustrating the overall method of pixel-based network routing, showing multiple data streams being multiplexed into a single frame based on color.

**FIG. 2** is a block diagram showing the system architecture, highlighting the role of the processor as a software-defined switch.

**FIG. 3** is a flowchart of the pixel-based routing process, from packet analysis to frame generation and final decoding.

## Detailed Description of the Invention

Referring to **FIG. 1**, the method utilizes a massively parallel optical interconnect to multiplex several incoming data streams (101, 102, 103), each intended for a different logical destination (e.g., "Network A", "Network B", "Network C"). A processor, acting as a Software-Defined Switch (104), analyzes the data from each stream.

The switch (104) maintains a routing table that maps each destination to a specific emission characteristic. In this embodiment, Network A is mapped to the color Red, Network B to Green, and Network C to Blue. The switch (104) generates a composite data frame (105) for the optical interconnect's emitter array. Within this frame, data from the "Network A" stream is placed into pixels that are set to red (106). Data from the "Network B" stream is placed into pixels that are set to green (107), and data from the "Network C" stream is placed into pixels that are set to blue (108).

The entire frame (105), containing a heterogeneous mix of colored pixels, is transmitted as a single, parallel unit. At the receiving end, a decoder (not shown) reads the color of each pixel and sorts the data accordingly, effectively demultiplexing the streams.

**FIG. 2** illustrates the system architecture. A host system provides multiple data streams (201, 202) to the routing system. A Processor/Switch (203) performs the core logic of the invention. It analyzes the incoming data, consults a Routing Table (204), and instructs an Encoder (205) on how to build the frame. The Encoder (205) generates the frame and sends it to the Emitter Array (206) for optical transmission. The signal is captured by a Sensor Array (207) and passed to a Decoder (208). The Decoder (208), aware of the routing-to-color mapping, reads the emission characteristic of each pixel and forwards the data to the correct output channels (209, 210), completing the routing process.

**FIG. 3** shows a flowchart of the process. The process begins with the system receiving data packets from multiple streams (301). For each packet, the processor determines its destination network (302). It then consults the routing table to find the corresponding emission characteristic (e.g., color) for that destination (303). The processor assigns the packet's data to an available pixel in the next frame to be transmitted and sets that pixel's color to the determined characteristic (304). This continues until the frame is full or a timing interval has passed (305). The completed frame is then transmitted optically (306). At the receiver, the frame is captured (307), and for each pixel, the data is decoded and routed to its final destination based on the pixel's color (308).

# Claims

What is claimed is:

1. A method for routing data over a massively parallel optical interconnect having a two-dimensional array of emission channels, the method comprising the steps of:

    - receiving a plurality of data packets, wherein each data packet is associated with a network destination from a set of possible destinations;
    - assigning a unique emission characteristic to each network destination in the set;
    - for each data packet, determining its associated network destination and identifying the corresponding unique emission characteristic;

- generating a data frame by assigning the data from said data packets to a plurality of channels in the two-dimensional array and setting the emission characteristic of each channel to the characteristic corresponding to the destination of the data it contains; and
- transmitting said data frame via the optical interconnect.

2. The method of claim 1, wherein the emission channels are pixels and the emission characteristic is the color of the pixels.

3. The method of claim 2, wherein the set of possible destinations is mapped to a set of colors, and wherein the step of generating the data frame comprises coloring the pixels according to the destination of the data they carry.

4. The method of claim 1, further comprising the steps of:

   - receiving, at a receiver, the transmitted data frame;
   - for each channel in the received frame, identifying its emission characteristic; and
   - routing the data from each channel to its network destination based on the identified emission characteristic.

5. The method of claim 1, wherein the steps are performed by a processor integrated into a connector housing of the optical interconnect.

6. A system for network routing, comprising:

   - a massively parallel optical interconnect including a two-dimensional array of individually addressable emission channels;
   - a processor configured to:
     - receive data streams associated with a plurality of network destinations;
     - maintain a routing table that maps each network destination to a unique emission characteristic;
     - generate a sequence of data frames by populating the channels of the array with data from the data streams and setting the emission characteristic of each channel according to the mapping in the routing table; and
   - an encoder circuit configured to drive the emission array to transmit the sequence of data frames.

7. The system of claim 6, wherein the emission channels are pixels on a micro-display and the unique emission characteristic is color.

8. The system of claim 6, further comprising a decoder circuit at a receiving end of the interconnect, the decoder configured to read the emission characteristic of each channel and to route the data contained therein to the corresponding network destination.

9. The system of claim 6, wherein the processor is a GPU, FPGA, or ASIC located within a connector head of the interconnect.

10. The system of claim 6, wherein the routing method constitutes a form of software-defined wavelength-division multiplexing.

# Drawings

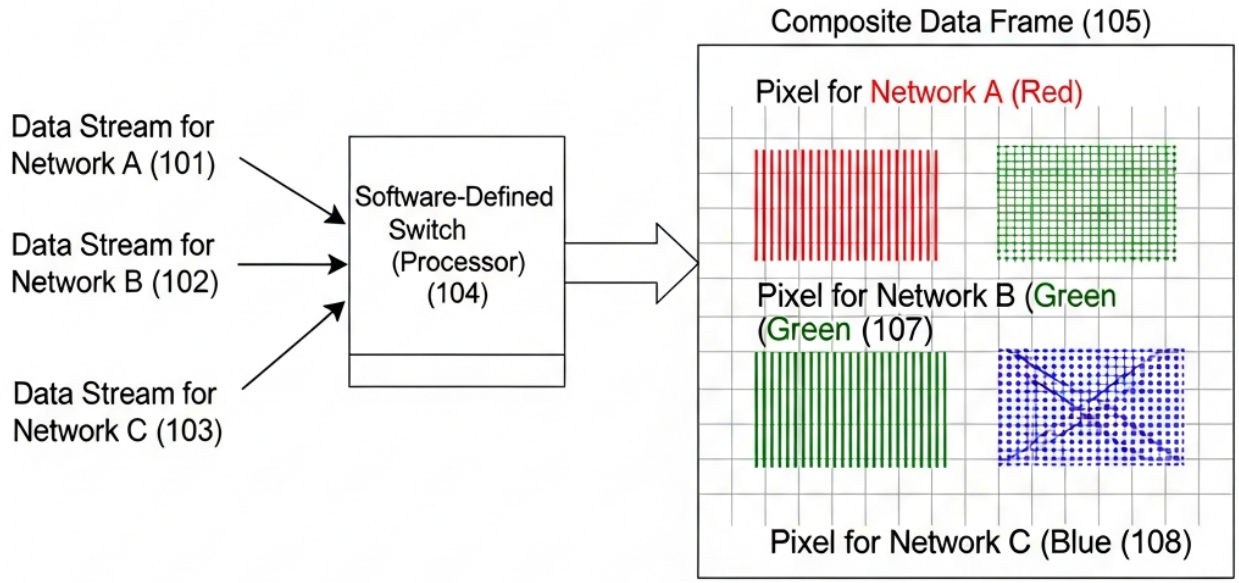



Figure 1: A conceptual diagram illustrating the overall method of pixel-based network routing, showing multiple data streams being multiplexed into a single frame based on color.

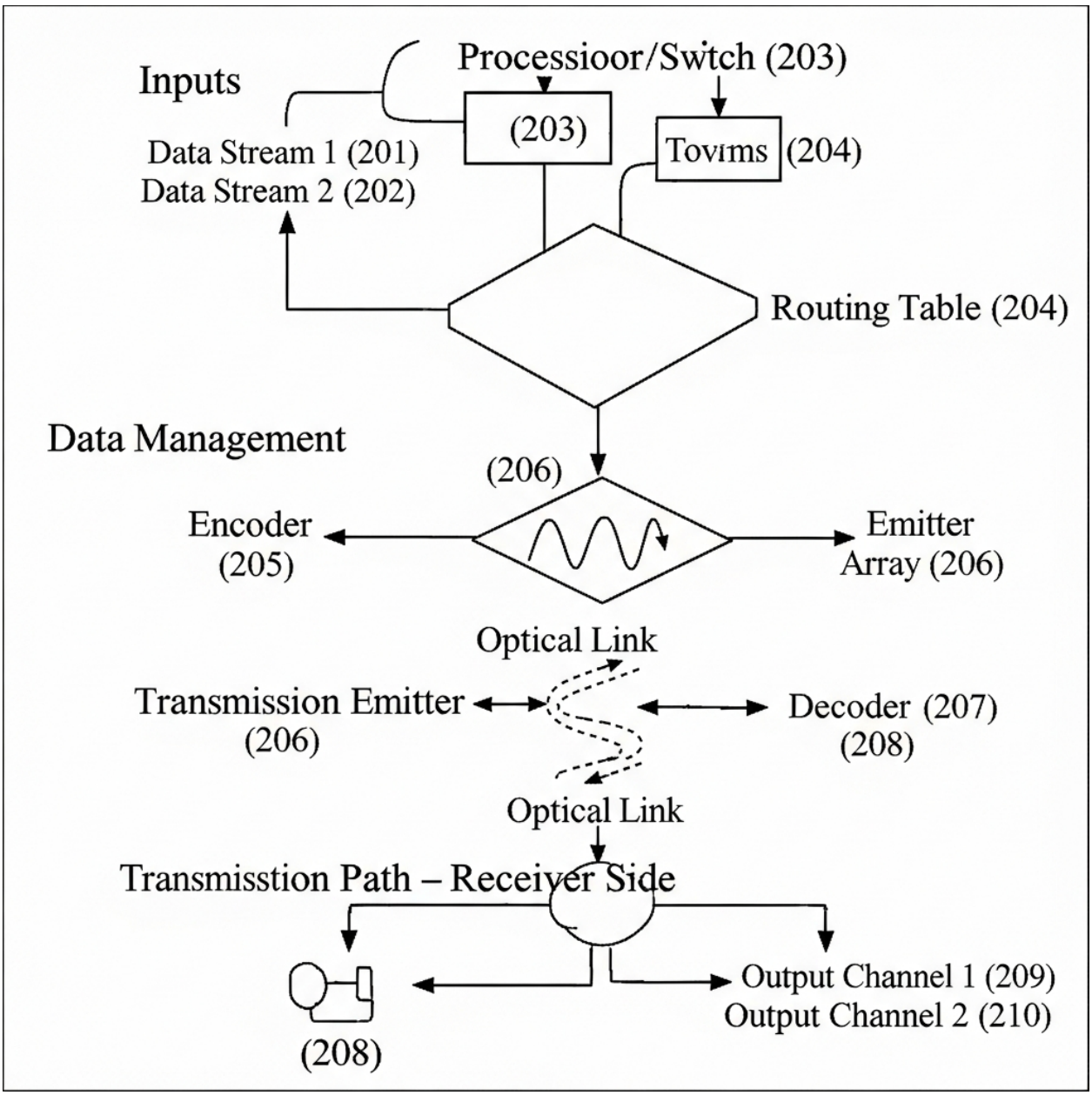



Figure 2: A block diagram showing the system architecture, highlighting the role of the processor as a software-defined switch.

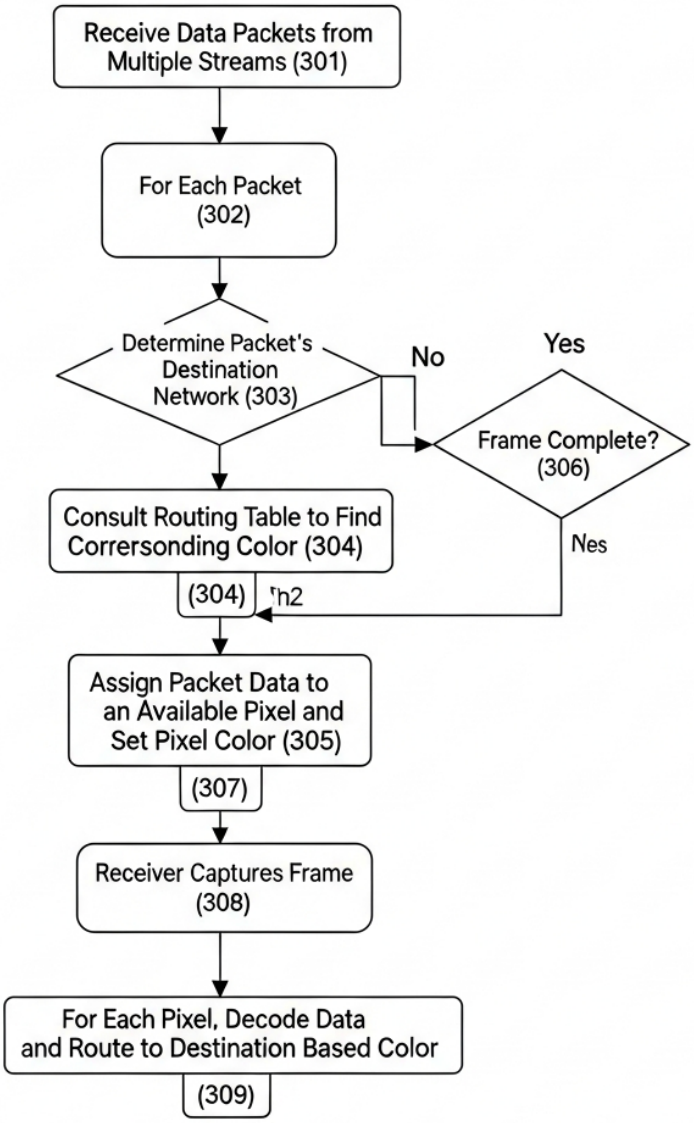



Figure 3: A flowchart of the pixel-based routing process, from packet analysis to frame generation and final decoding.